EN 18222 ReadDPPById 接口、Full/Compressed 表示与欧盟电池护照法规字段

Table of Contents

EN 18222 ReadDPPById 接口、Full/Compressed 表示与欧盟电池护照法规字段

版本：1.2（法规数据点中英文对照版）
整理日期：2026-08-27
适用对象：DPP 平台产品、架构、后端、前端和接口联调人员

1. 结论先行

1. `ReadDPPById` 是 HTTP GET 读取接口，请求没有 JSON Body。
2. `full` 和 `compressed` 通过查询参数 `representation` 选择，描述的是响应中 DPP 的表示方式。
3. 欧盟已经公布电池护照的 71 个法规业务数据点及法律来源。
4. 欧盟尚未完全公布可直接用于编码的最终电池语义目录，包括统一的 `elementId`、正式 `dictionaryReference` URI、所有单位、精度、格式和枚举。
5. 本文中的驼峰字段名、`.example.com` URI 和业务数值属于可开发的临时实施映射，不是欧盟官方机器字段。

2. 法规及资料来源

- Regulation (EU) 2023/1542，尤其是 Article 77、Annex VI 和 Annex XIII。
- European Commission，《Digital Batteries Passport – data points by category》，Version 2.0，2026-08-15。
- EN 18222:2026：DPP API 逻辑方法和接口行为。
- EN 18223:2026：DPP 数据模型以及 full/compressed 表示结构。

官方资料：

- 欧委会电池护照 71 项数据指南：https://single-market-economy.ec.europa.eu/document/download/cd1e5e6c-4a4a-4b99-995a-49eb6916187e_en?filename=Digital+Batteries+Passport+-+data+point+by+category.pdf

- 欧委会发布说明：https://single-market-economy.ec.europa.eu/news/guidance-support-preparations-digital-batteries-passport-2026-08-21_en
- Regulation (EU) 2023/1542 合并文本：https://eur-lex.europa.eu/eli/reg/2023/1542/2025-07-31/eng/pdf
- 欧盟 DPP Registry：https://single-market-economy.ec.europa.eu/single-market/digital-product-passport/dpp-registry_en

3. 欧盟电池护照 71 个真实法规数据点

英文名称逐项采用欧委会 Version 2.0 指南。中文列为便于研发理解的工程翻译，不代替英文法规原文。

编号	Data point name (official English)	中文对照（工程翻译）	法律来源	2027-02 起适用说明
1	Unique identifier	唯一标识符	BR Article 77(3)	EV、LMT、工业电池必填
2	Identity of who is registering and/or is responsible for the battery passport	注册和/或负责电池护照的主体身份	BR Article 77(3)	全类别必填
3	Manufacturer name, registered trade name or registered trademark	制造商名称、注册商号或注册商标	BR Annex VI A(1)	全类别必填
4	Manufacturer postal address, indicating a single contact point	制造商邮政地址及单一联系点	BR Annex VI A(1)	全类别必填
5	If available, manufacturer web and email address	制造商网站和电子邮箱（如有）	BR Annex VI A(1)	信息存在时填写
6	Battery category	电池类别	BR Annex VI A(2)	全类别必填
7	Model identification and batch or serial number, or product number or another element allowing their identification	型号及批次号、序列号、产品号或其他识别信息	BR Annex VI A(2)	全类别必填
8	The place of manufacturer (geographical location of a battery manufacturing plant)	制造地点（电池制造厂的地理位置）	BR Annex VI A(3)	全类别必填
9	The date of manufacturing (month and year)	制造日期（月和年）	BR Annex VI A(4)	全类别必填
10	The weight	重量	BR Annex	全类别必填

编号	Data point name (official English)	中文对照（工程翻译）	法律来源	2027-02 起适用说明
			VI A(5)	
11	The capacity	容量	BR Annex VI A(6)	全类别必填
12	The chemistry	化学体系	BR Annex VI A(7)	全类别必填
13	The hazardous substances present in the battery, other than mercury, cadmium or lead	电池内除汞、镉、铅以外的有害物质	BR Annex VI A(8)	全类别必填
14	Usable extinguishing agent	可使用的灭火剂	BR Annex VI A(9)	全类别必填
15	Critical raw materials present in the battery in a concentration of more than 0,1 % weight by weight	电池中质量浓度超过 0.1% 的关键原材料	BR Annex VI A(10)	全类别必填
16	The material composition of the battery, including its chemistry, hazardous substances present in the battery, other than mercury, cadmium or lead, and critical raw materials present in the battery	电池材料组成，包括化学体系、有害物质和关键原材料	BR Annex XIII 1(b)	与前述字段重复，暂不单独填写/展示
17	The carbon footprint declaration	碳足迹声明	BR Annex XIII 1(c)	截至 2027-02 格式尚待实施法案规定
18	The carbon footprint label	碳足迹标签	BR Annex XIII 1(c)	截至 2027-02 格式尚待实施法案规定
19	Information on responsible sourcing as indicated in the report on battery due diligence policy referred to in Article 52(3)	电池尽职调查政策报告中的负责任采购信息	BR Annex XIII 1(d)	预计随 Article 48 从 2027-08 起适用
20	Percentage share of cobalt that is present in active materials and that has been recovered from battery manufacturing waste or post-consumer waste	活性材料中来自制造废料或消费后废弃物回收的钴占比	BR Annex XIII 1(e)	依 Article 8 及相关授权法案实施
21	Percentage share of lithium that is present in active materials and that has been recovered from battery manufacturing waste or post-consumer waste	活性材料中来自制造废料或消费后废弃物回收的锂占比	BR Annex XIII 1(e)	依 Article 8 及相关授权法案实施

编号	Data point name (official English)	中文对照（工程翻译）	法律来源	2027-02 起适用说明
22	Percentage share of nickel that is present in active materials and that has been recovered from battery manufacturing waste or post-consumer waste	活性材料中来自制造废料或消费后废弃物回收的镍占比	BR Annex XIII 1(e)	依 Article 8 及相关授权法案实施
23	The percentage share of lead that is present in the battery and that has been recovered from waste	电池中来自废弃物回收的铅占比	BR Annex XIII 1(e)	依 Article 8 及相关授权法案实施
24	The share of renewable content	可再生来源内容物占比	BR Annex XIII 1(f)	全类别必填
25	Rated capacity (in Ah)	额定容量（Ah）	BR Annex XIII 1(g)	与数据点 11 重复，暂不重复展示
26	Minimal voltage, with temperature range when relevant	最低电压，以及相关时对应的温度范围	BR Annex XIII 1(h)	全类别必填
27	Nominal voltage, with temperature range when relevant	标称电压，以及相关时对应的温度范围	BR Annex XIII 1(h)	全类别必填
28	Maximum voltage, with temperature range when relevant	最高电压，以及相关时对应的温度范围	BR Annex XIII 1(h)	全类别必填
29	Original power capability (in Watts)	初始功率能力（W）	BR Annex XIII 1(i)	全类别必填
30	Power limits, with temperature range when relevant	功率限制，以及相关时对应的温度范围	BR Annex XIII 1(i)	全类别必填
31	Expected battery lifetime expressed in cycles	以循环次数表示的预期电池寿命	BR Annex XIII 1(j)	EV、LMT 必填；部分工业电池适用
32	Reference test used for expected battery lifetime expressed in cycles	预期循环寿命采用的参考测试	BR Annex XIII 1(j)	EV、LMT 必填；部分工业电池适用
33	Capacity threshold for exhaustion	判定电池耗尽的容量阈值	BR Annex XIII 1(k)	EV 必填；LMT、工业电池暂不展示
34	Temperature range the battery can withstand when not in use (reference test)	电池停用时可承受的温度范围（参考测试）	BR Annex XIII 1(l)	全类别必填

编号	Data point name (official English)	中文对照（工程翻译）	法律来源	2027-02 起适用说明
35	Period for which the commercial warranty for the calendar life applies	商业质保适用的日历寿命期限	BR Annex XIII 1(m)	存在商业质保时填写
36	Initial round trip energy efficiency	初始往返能量效率	BR Annex XIII 1(n)	EV、LMT 必填；部分工业电池适用
37	Round trip energy efficiency at 50 % of cycle-life	循环寿命达到 50% 时的往返能量效率	BR Annex XIII 1(n)	EV、LMT 必填；部分工业电池适用
38	Internal battery cell and pack resistance	电芯和电池包内部电阻	BR Annex XIII 1(o)	全类别必填
39	C-rate of relevant cycle-life test	相关循环寿命测试采用的 C 倍率	BR Annex XIII 1(p)	EV、LMT 必填；部分工业电池适用
40	The marking requirements laid down in Article 13(4)	Article 13(4) 规定的标识信息	BR Annex XIII 1(q)	全类别必填
41	The marking requirements laid down in Article 13(5)	Article 13(5) 规定的标识信息	BR Annex XIII 1(q)	含镉或铅且达到适用条件时填写
42	The EU declaration of conformity referred to in Article 18	Article 18 所述欧盟符合性声明	BR Annex XIII 1(r)	全类别必填
43	The information regarding the prevention and management of waste batteries laid down in Article 74(1), points (a) to (f).	Article 74(1)(a)-(f) 的废电池预防和管理信息	BR Annex XIII 1(s)	全类别必填
44	Clear, understandable and readable instructions for use in a format that makes it possible to print, download and save them on an electronic device so that the user can access them at all times, in particular during a breakdown of the battery	可打印、下载、保存并可随时访问的清晰易懂使用说明，尤其供电池故障时使用	BR Annex XIII 1(t)	截至 2027-02 等待 Omnibus 后续规定
45	Detailed composition, including materials used in the cathode, anode and electrolyte	详细组成，包括阴极、阳极和电解质所用材料	BR Annex XIII 2(a)	全类别必填
46	Part numbers for components	组件零件编号	BR Annex XIII 2(b)	全类别必填
47	Contact details of sources for replacement spares	替换备件来源的联系信息	BR Annex	全类别必填

编号	Data point name (official English)	中文对照（工程翻译）	法律来源	2027-02 起适用说明
			XIII 2(b)	
48	Dismantling information, including at least: exploded diagrams of the battery system/pack showing the location of battery cells; disassembly sequences; type and number of fastening techniques to be unlocked; tools required for disassembly; warnings if risk of damaging parts exist; amount of cells used and layout	拆解信息，包括爆炸图、拆卸顺序、紧固方式、工具、损坏风险警告、电芯数量和布局等	BR Annex XIII 2(c)	全类别必填
49	Safety measures	安全措施	BR Annex XIII 2(d)	全类别必填
50	Results of test reports proving compliance with the requirements laid down in this Regulation or any delegated or implementing act adopted pursuant to this Regulation	证明符合法规及相关授权或实施法案要求的测试报告结果	BR Annex XIII 3	全类别必填
51	Rated capacity (in Ah)	动态额定容量（Ah）	BR Annex XIII 4(a)	EV、LMT 为数据点 11 的动态值；工业电池按适用性
52	Capacity fade (in %)	容量衰减（%）	BR Annex XIII 4(a)	EV、LMT 必填；工业电池按适用性
53	Power (in W)	当前功率（W）	BR Annex XIII 4(a)	EV、LMT 必填；工业电池按适用性
54	Power fade (in %)	功率衰减（%）	BR Annex XIII 4(a)	EV、LMT 必填；工业电池按适用性
55	Internal resistance (in Ω)	内部电阻（Ω）	BR Annex XIII 4(a)	EV、LMT 必填；工业电池按适用性
56	Internal resistance increase (in %)	内部电阻增长（%）	BR Annex XIII 4(a)	EV、LMT 必填；工业电池按适用性
57	Where applicable, energy round trip efficiency (in %)	适用时的能量往返效率（%）	BR Annex XIII 4(a)	全类别按适用性

编号	Data point name (official English)	中文对照（工程翻译）	法律来源	2027-02 起适用说明
58	Where applicable, energy round trip efficiency fade (in %)	适用时的能量往返效率衰减（%）	BR Annex XIII 4(a)	全类别按适用性
59	The expected life-time of the battery under the reference conditions for which it has been designed, in terms of cycles, except for non-cycle applications	设计参考条件下以循环次数表示的预期寿命，非循环应用除外	BR Annex XIII 4(a)	EV、LMT 必填；工业电池按适用性
60	The expected life-time of the battery under the reference conditions for which it has been designed, in terms of calendar years	设计参考条件下以日历年表示的预期寿命	BR Annex XIII 4(a)	EV、LMT 必填；工业电池按适用性
61	Information on the state of health of the battery pursuant to Article 14: state of certified energy (SOCE)	Article 14 电池健康状态：认证能量状态（SOCE）	BR Annex XIII 4(b)	EV 必填；LMT、工业电池暂不展示
62	Information on the state of health of the battery pursuant to Article 14: remaining capacity	Article 14 电池健康状态：剩余容量	BR Annex XIII 4(b)	LMT 必填；工业电池按适用性；EV 暂不展示
63	Information on the state of health of the battery pursuant to Article 14: where possible, the remaining power capability	Article 14 电池健康状态：可行时的剩余功率能力	BR Annex XIII 4(b)	LMT 必填；工业电池按适用性；EV 暂不展示
64	Information on the state of health of the battery pursuant to Article 14: where possible, the remaining round trip efficiency	Article 14 电池健康状态：可行时的剩余往返效率	BR Annex XIII 4(b)	LMT 必填；工业电池按适用性；EV 暂不展示
65	Information on the state of health of the battery pursuant to Article 14: [where possible,] the evolution of self-discharging rates	Article 14 电池健康状态：可行时的自放电率变化	BR Annex XIII 4(b)	LMT 必填；工业电池按适用性；EV 暂不展示
66	Information on the state of health of the battery pursuant to Article 14: where possible, the ohmic resistance	Article 14 电池健康状态：可行时的欧姆内阻	BR Annex XIII 4(b)	LMT 必填；工业电池按适用性；EV 暂不展示
67	Information on the status of the battery, defined as ‘original’, ‘repurposed’, ‘re-used’, ‘remanufactured’ or ‘waste’;	电池状态：原始、改变用途、再使用、再制造或废弃	BR Annex XIII 4(c)	全类别必填

编号	Data point name (official English)	中文对照（工程翻译）	法律来源	2027-02 起适用说明
68	The number of charging and discharging cycles	充放电循环次数	BR Annex XIII 4(d)	全类别按适用性
69	Negative events, such as accidents	事故等负面事件	BR Annex XIII 4(d)	全类别按适用性
70	Periodically recorded information on the operating environmental conditions, including temperature	周期记录的运行环境条件，包括温度	BR Annex XIII 4(d)	全类别按适用性
71	Periodically recorded information on the state of charge	周期记录的荷电状态（SOC）	BR Annex XIII 4(d)	全类别按适用性

4. ReadDPPById 的真实 HTTP 入参

接口：

```
GET /v1/dpps/{dppId}?representation={full|compressed}
```

参数：

位置	参数	类型	必需	说明
Path	`dppId`	string / DppIdentifier	是	DPP 唯一标识，放入 URL 路径前执行 UTF-8 百分号编码
Query	`representation`	enum	否	`full` 或 `compressed`；本实施配置默认 `compressed`
Header	`Accept`	string	建议	`application/json`
Header	`Authorization`	string	视权限	`Bearer <JWT>`，公开字段可允许匿名读取
Header	`X-Correlation-ID`	string	否	调用链跟踪标识
Body	无	—	—	GET 接口不传 JSON 请求体

5. Full 请求参数 JSON

以下 JSON 是为了在 Apifox、Java DTO 或自动化测试中统一表达 Path、Query、Header 参数，不是发送给服务端的 **HTTP Body**。

```
{
  "pathParameters": {
    "dppId": "urn:uuid:0198f87e-7c2a-7c58-91a3-3b83f6e91240"
  },
  "queryParameters": {
    "representation": "full"
  },
  "headers": {
    "Accept": "application/json",
    "Authorization": "Bearer <JWT>",
    "X-Correlation-ID": "01K3QBTB8F3CBTDPND6GHRFQAG"
  }
}
```

实际请求：

```
GET /v1/dpps/urn%3Auuid%3A0198f87e-7c2a-7c58-91a3-3b83f6e91240?representation=full HTTP/1.1
Host: dpp.example.com
Accept: application/json
Authorization: Bearer <JWT>
X-Correlation-ID: 01K3QBTB8F3CBTDPND6GHRFQAG
```

6. Compressed 请求参数 JSON

```
{
  "pathParameters": {
    "dppId": "urn:uuid:0198f87e-7c2a-7c58-91a3-3b83f6e91240"
  },
  "queryParameters": {
    "representation": "compressed"
  },
  "headers": {
    "Accept": "application/json",
    "Authorization": "Bearer <JWT>",
    "X-Correlation-ID": "01K3QBTB8F3CBTDPND6GHRFQAG"
  }
}
```

实际请求：

```
GET /v1/dpps/urn%3Auuid%3A0198f87e-7c2a-7c58-91a3-3b83f6e91240?representation=compressed HTTP/1.1
Host: dpp.example.com
```

```
Accept: application/json
Authorization: Bearer <JWT>
X-Correlation-ID: 01K3QBTB8F3CBTDPND6GHRFQAG
```

7. Full 响应 JSON 示例

示例产品是一块 48 V、20 Ah 的 LMT 电动自行车电池。业务数据项来自法规清单，数值均为模拟值。`elementId` 和 `dictionaryReference` 是临时实施映射。


```json
{
  "statusCode": "Success",
  "payload": {
    "digitalProductPassportId": "urn:uuid:0198f87e-7c2a-7c58-91a3-3b83f6e91240",
    "uniqueProductIdentifier": "https://id.gs1.org/01/09506000134352/21/LMT26B00018427",
    "granularity": "item",
    "dppSchemaVersion": "company:battery-passport:0.9.0",
    "dppStatus": "active",
    "lastUpdated": "2026-08-27T08:42:17Z",
    "economicOperatorId": "iso6523-actorid-upis::0088:9506000134352",
    "facilityId": "iso6523-actorid-upis::0088:95060001343520001",
    "contentSpecificationIds": [
      "EU-2023-1542:Annex-VI",
      "EU-2023-1542:Annex-XIII",
      "EN-18223:2026"
    ],
    "elements": [
      {
        "elementId": "batteryIdentification",
        "objectType": "DataElementCollection",
        "dictionaryReference": "https://vocabulary.example.com/battery/identification",
        "elements": [
          {
            "elementId": "manufacturerName",
            "objectType": "SingleValuedDataElement",
            "dictionaryReference": "https://vocabulary.example.com/battery/dp03-manufacturer-name",
            "valueDataType": "xsd:string",
            "value": "Example Energy Systems GmbH"
          },
          {
            "elementId": "manufacturerPostalAddress",
```

```
    "objectType": "MultiLanguageDataElement",
    "dictionaryReference": "https://vocabulary.example.com/battery/dp04-manufacturer-postal-address",
    "value": [
      {
        "language": "en",
        "value": "Industriestrasse 18, 70565 Stuttgart, Germany"
      },
      {
        "language": "de",
        "value": "Industriestraße 18, 70565 Stuttgart, Deutschland"
      }
    ]
  },
  {
    "elementId": "manufacturerWebsite",
    "objectType": "SingleValuedDataElement",
    "dictionaryReference": "https://vocabulary.example.com/battery/dp05-manufacturer-website",
    "valueDataType": "xsd:anyURI",
    "value": "https://manufacturer.example.com"
  },
  {
    "elementId": "batteryCategory",
    "objectType": "SingleValuedDataElement",
    "dictionaryReference": "https://vocabulary.example.com/battery/dp06-battery-category",
    "valueDataType": "xsd:string",
    "value": "LMT"
  },
  {
    "elementId": "modelIdentification",
    "objectType": "SingleValuedDataElement",
    "dictionaryReference": "https://vocabulary.example.com/battery/dp07-model-identification",
    "valueDataType": "xsd:string",
    "value": "EBP-48V-20AH-PRO"
  },
  {
    "elementId": "serialNumber",
    "objectType": "SingleValuedDataElement",
    "dictionaryReference": "https://vocabulary.example.com/battery/dp07-serial-number",
    "valueDataType": "xsd:string",
    "value": "LMT26B00018427"
```

```
    },
    {
      "elementId": "placeOfManufacture",
      "objectType": "SingleValuedDataElement",
      "dictionaryReference": "https://vocabulary.example.com/battery/dp08-place-of-manufacture",
      "valueDataType": "xsd:string",
      "value": "Stuttgart, Baden-Württemberg, Germany"
    },
    {
      "elementId": "dateOfManufacture",
      "objectType": "SingleValuedDataElement",
      "dictionaryReference": "https://vocabulary.example.com/battery/dp09-date-of-manufacture",
      "valueDataType": "xsd:string",
      "value": "2026-06"
    }
  ]
},
{
  "elementId": "technicalCharacteristics",
  "objectType": "DataElementCollection",
  "dictionaryReference": "https://vocabulary.example.com/battery/technical-characteristics",
  "elements": [
    {
      "elementId": "weight",
      "objectType": "SingleValuedDataElement",
      "dictionaryReference": "https://vocabulary.example.com/battery/dp10-weight-kg",
      "valueDataType": "xsd:decimal",
      "value": 5.84
    },
    {
      "elementId": "capacity",
      "objectType": "SingleValuedDataElement",
      "dictionaryReference": "https://vocabulary.example.com/battery/dp11-capacity-ah",
      "valueDataType": "xsd:decimal",
      "value": 20.0
    },
    {
      "elementId": "chemistry",
      "objectType": "SingleValuedDataElement",
      "dictionaryReference": "https://vocabulary.example.com/battery/dp12-chemistry",
```

```
    "valueDataType": "xsd:string",
    "value": "Lithium-ion NMC 811 / Graphite"
  },
  {
    "elementId": "minimalVoltage",
    "objectType": "SingleValuedDataElement",
    "dictionaryReference": "https://vocabulary.example.com/battery/dp26-minimal-voltage-v",
    "valueDataType": "xsd:decimal",
    "value": 39.0
  },
  {
    "elementId": "nominalVoltage",
    "objectType": "SingleValuedDataElement",
    "dictionaryReference": "https://vocabulary.example.com/battery/dp27-nominal-voltage-v",
    "valueDataType": "xsd:decimal",
    "value": 48.0
  },
  {
    "elementId": "maximumVoltage",
    "objectType": "SingleValuedDataElement",
    "dictionaryReference": "https://vocabulary.example.com/battery/dp28-maximum-voltage-v",
    "valueDataType": "xsd:decimal",
    "value": 54.6
  },
  {
    "elementId": "originalPowerCapability",
    "objectType": "SingleValuedDataElement",
    "dictionaryReference": "https://vocabulary.example.com/battery/dp29-original-power-capability-w",
    "valueDataType": "xsd:decimal",
    "value": 1000.0
  },
  {
    "elementId": "expectedBatteryLifetimeCycles",
    "objectType": "SingleValuedDataElement",
    "dictionaryReference": "https://vocabulary.example.com/battery/dp31-expected-lifetime-cycles",
    "valueDataType": "xsd:integer",
    "value": 1000
  },
  {
    "elementId": "storageTemperatureRange",
```

```
            "objectType": "SingleValuedDataElement",
            "dictionaryReference": "https://vocabulary.example.com/battery/dp34-storage-temperature-range",
            "valueDataType": "xsd:string",
            "value": "-20 °C to +45 °C"
          },
          {
            "elementId": "initialRoundTripEnergyEfficiency",
            "objectType": "SingleValuedDataElement",
            "dictionaryReference": "https://vocabulary.example.com/battery/dp36-initial-round-trip-efficiency-
percent",
            "valueDataType": "xsd:decimal",
            "value": 94.2
          },
          {
            "elementId": "internalBatteryResistance",
            "objectType": "SingleValuedDataElement",
            "dictionaryReference": "https://vocabulary.example.com/battery/dp38-pack-resistance-ohm",
            "valueDataType": "xsd:decimal",
            "value": 0.1
          }
        ]
      },
      {
        "elementId": "materialComposition",
        "objectType": "DataElementCollection",
        "dictionaryReference": "https://vocabulary.example.com/battery/material-composition",
        "elements": [
          {
            "elementId": "hazardousSubstances",
            "objectType": "MultiValuedDataElement",
            "dictionaryReference": "https://vocabulary.example.com/battery/dp13-hazardous-substances",
            "valueDataType": "xsd:string",
            "value": [
              {
                "elementId": "hazardousSubstance01",
                "objectType": "SingleValuedDataElement",
                "valueDataType": "xsd:string",
                "value": "1-methyl-2-pyrrolidone (NMP), CAS 872-50-4"
              },
              {
```

```
      "elementId": "hazardousSubstance02",
      "objectType": "SingleValuedDataElement",
      "valueDataType": "xsd:string",
      "value": "Lithium hexafluorophosphate, CAS 21324-40-3"
    }
  ]
},
{
  "elementId": "criticalRawMaterials",
  "objectType": "MultiValuedDataElement",
  "dictionaryReference": "https://vocabulary.example.com/battery/dp15-critical-raw-materials",
  "valueDataType": "xsd:string",
  "value": [
    {
      "elementId": "criticalRawMaterial01",
      "objectType": "SingleValuedDataElement",
      "valueDataType": "xsd:string",
      "value": "Lithium"
    },
    {
      "elementId": "criticalRawMaterial02",
      "objectType": "SingleValuedDataElement",
      "valueDataType": "xsd:string",
      "value": "Cobalt"
    },
    {
      "elementId": "criticalRawMaterial03",
      "objectType": "SingleValuedDataElement",
      "valueDataType": "xsd:string",
      "value": "Natural graphite"
    }
  ]
},
{
  "elementId": "detailedComposition",
  "objectType": "DataElementCollection",
  "dictionaryReference": "https://vocabulary.example.com/battery/dp45-detailed-composition",
  "elements": [
    {
      "elementId": "cathodeMaterial",
```

```
          "objectType": "SingleValuedDataElement",
          "valueDataType": "xsd:string",
          "value": "LiNi0.8Mn0.1Co0.1O2"
        },
        {
          "elementId": "anodeMaterial",
          "objectType": "SingleValuedDataElement",
          "valueDataType": "xsd:string",
          "value": "Graphite with 5% silicon composite"
        },
        {
          "elementId": "electrolyte",
          "objectType": "SingleValuedDataElement",
          "valueDataType": "xsd:string",
          "value": "LiPF6 in EC/EMC electrolyte"
        },
        {
          "elementId": "numberOfCells",
          "objectType": "SingleValuedDataElement",
          "valueDataType": "xsd:integer",
          "value": 52
        },
        {
          "elementId": "cellLayout",
          "objectType": "SingleValuedDataElement",
          "valueDataType": "xsd:string",
          "value": "13S4P"
        }
      ]
    }
  ]
},
{
  "elementId": "documentsAndSafety",
  "objectType": "DataElementCollection",
  "dictionaryReference": "https://vocabulary.example.com/battery/documents-and-safety",
  "elements": [
    {
      "elementId": "usableExtinguishingAgent",
      "objectType": "MultiLanguageDataElement",
```

```
      "dictionaryReference": "https://vocabulary.example.com/battery/dp14-usable-extinguishing-agent",
      "value": [
        {
          "language": "en",
          "value": "Use large quantities of water for cooling; avoid enclosed-space exposure to vapours."
        },
        {
          "language": "de",
          "value": "Mit großen Mengen Wasser kühlen; Dämpfe in geschlossenen Räumen vermeiden."
        }
      ]
    },
    {
      "elementId": "euDeclarationOfConformity",
      "objectType": "RelatedResource",
      "dictionaryReference": "https://vocabulary.example.com/battery/dp42-eu-declaration-of-conformity",
      "contentType": "application/pdf",
      "url": "https://documents.example.com/batteries/LMT26B00018427/eu-declaration.pdf",
      "language": "en",
      "resourceTitle": "EU Declaration of Conformity"
    },
    {
      "elementId": "dismantlingInformation",
      "objectType": "RelatedResource",
      "dictionaryReference": "https://vocabulary.example.com/battery/dp48-dismantling-information",
      "contentType": "application/pdf",
      "url": "https://documents.example.com/batteries/LMT26B00018427/dismantling.pdf",
      "language": "en",
      "resourceTitle": "Battery pack dismantling instructions and exploded diagram"
    },
    {
      "elementId": "safetyMeasures",
      "objectType": "MultiLanguageDataElement",
      "dictionaryReference": "https://vocabulary.example.com/battery/dp49-safety-measures",
      "value": [
        {
          "language": "en",
          "value": "Disconnect the battery before servicing. Do not crush, puncture, short-circuit or expose
above 60 °C."
        },
```

```
              {
                "language": "de",
                "value": "Vor Wartungsarbeiten Batterie trennen. Nicht zerdrücken, durchstechen, kurzschließen oder
über 60 °C erhitzen."
              }
            ]
          },
          {
            "elementId": "complianceTestReport",
            "objectType": "RelatedResource",
            "dictionaryReference": "https://vocabulary.example.com/battery/dp50-test-reports",
            "contentType": "application/pdf",
            "url": "https://documents.example.com/batteries/LMT26B00018427/test-report.pdf",
            "language": "en",
            "resourceTitle": "Performance, durability and safety test report"
          }
        ]
      },
      {
        "elementId": "currentPerformanceAndStateOfHealth",
        "objectType": "DataElementCollection",
        "dictionaryReference": "https://vocabulary.example.com/battery/current-performance",
        "elements": [
          {
            "elementId": "ratedCapacity",
            "objectType": "SingleValuedDataElement",
            "dictionaryReference": "https://vocabulary.example.com/battery/dp51-rated-capacity-ah",
            "valueDataType": "xsd:decimal",
            "value": 18.6
          },
          {
            "elementId": "capacityFade",
            "objectType": "SingleValuedDataElement",
            "dictionaryReference": "https://vocabulary.example.com/battery/dp52-capacity-fade-percent",
            "valueDataType": "xsd:decimal",
            "value": 7.0
          },
          {
            "elementId": "power",
            "objectType": "SingleValuedDataElement",
```

```
            "dictionaryReference": "https://vocabulary.example.com/battery/dp53-power-w",
            "valueDataType": "xsd:decimal",
            "value": 925.0
          },
          {
            "elementId": "internalResistance",
            "objectType": "SingleValuedDataElement",
            "dictionaryReference": "https://vocabulary.example.com/battery/dp55-internal-resistance-ohm",
            "valueDataType": "xsd:decimal",
            "value": 0.118
          },
          {
            "elementId": "remainingCapacity",
            "objectType": "SingleValuedDataElement",
            "dictionaryReference": "https://vocabulary.example.com/battery/dp62-remaining-capacity-ah",
            "valueDataType": "xsd:decimal",
            "value": 18.6
          },
          {
            "elementId": "remainingPowerCapability",
            "objectType": "SingleValuedDataElement",
            "dictionaryReference": "https://vocabulary.example.com/battery/dp63-remaining-power-capability-w",
            "valueDataType": "xsd:decimal",
            "value": 925.0
          },
          {
            "elementId": "remainingRoundTripEfficiency",
            "objectType": "SingleValuedDataElement",
            "dictionaryReference": "https://vocabulary.example.com/battery/dp64-remaining-round-trip-efficiency-
percent",
            "valueDataType": "xsd:decimal",
            "value": 91.3
          },
          {
            "elementId": "batteryStatus",
            "objectType": "SingleValuedDataElement",
            "dictionaryReference": "https://vocabulary.example.com/battery/dp67-battery-status",
            "valueDataType": "xsd:string",
            "value": "original"
          },
```

```
          {
            "elementId": "chargingDischargingCycleCount",
            "objectType": "SingleValuedDataElement",
            "dictionaryReference": "https://vocabulary.example.com/battery/dp68-cycle-count",
            "valueDataType": "xsd:integer",
            "value": 327
          },
          {
            "elementId": "stateOfCharge",
            "objectType": "SingleValuedDataElement",
            "dictionaryReference": "https://vocabulary.example.com/battery/dp71-state-of-charge-percent",
            "valueDataType": "xsd:decimal",
            "value": 76.4
          },
          {
            "elementId": "lastMeasurementTime",
            "objectType": "SingleValuedDataElement",
            "dictionaryReference": "https://vocabulary.example.com/battery/measurement-time",
            "valueDataType": "xsd:dateTime",
            "value": "2026-08-27T08:40:00Z"
          }
        ]
      }
    ]
  }
}
```


8. Compressed 响应 JSON 示例

Compressed 表示省略 `objectType`、`dictionaryReference` 和 `valueDataType` 等重复语义元数据，直接用 `elementId` 作为业务 JSON Key。字段定义、类型和单位由版本化数据字典解释。


```
{
  "statusCode": "Success",
  "payload": {
    "digitalProductPassportId": "urn:uuid:0198f87e-7c2a-7c58-91a3-3b83f6e91240",
    "uniqueProductIdentifier": "https://id.gs1.org/01/09506000134352/21/LMT26B00018427",
    "granularity": "item",
    "dppSchemaVersion": "company:battery-passport:0.9.0",
    "dppStatus": "active",
```

```
"lastUpdated": "2026-08-27T08:42:17Z",
"economicOperatorId": "iso6523-actorid-upis::0088:9506000134352",
"facilityId": "iso6523-actorid-upis::0088:95060001343520001",
"contentSpecificationIds": [
  "EU-2023-1542:Annex-VI",
  "EU-2023-1542:Annex-XIII",
  "EN-18223:2026"
],
"batteryIdentification": {
  "manufacturerName": "Example Energy Systems GmbH",
  "manufacturerPostalAddress": {
    "en": "Industriestrasse 18, 70565 Stuttgart, Germany",
    "de": "Industriestraße 18, 70565 Stuttgart, Deutschland"
  },
  "manufacturerWebsite": "https://manufacturer.example.com",
  "batteryCategory": "LMT",
  "modelIdentification": "EBP-48V-20AH-PRO",
  "serialNumber": "LMT26B00018427",
  "placeOfManufacture": "Stuttgart, Baden-Württemberg, Germany",
  "dateOfManufacture": "2026-06"
},
"technicalCharacteristics": {
  "weight": 5.84,
  "capacity": 20.0,
  "chemistry": "Lithium-ion NMC 811 / Graphite",
  "minimalVoltage": 39.0,
  "nominalVoltage": 48.0,
  "maximumVoltage": 54.6,
  "originalPowerCapability": 1000.0,
  "expectedBatteryLifetimeCycles": 1000,
  "storageTemperatureRange": "-20 °C to +45 °C",
  "initialRoundTripEnergyEfficiency": 94.2,
  "internalBatteryResistance": 0.1
},
"materialComposition": {
  "hazardousSubstances": [
    "1-methyl-2-pyrrolidone (NMP), CAS 872-50-4",
    "Lithium hexafluorophosphate, CAS 21324-40-3"
  ],
  "criticalRawMaterials": [
```

```
        "Lithium",
        "Cobalt",
        "Natural graphite"
      ],
      "detailedComposition": {
        "cathodeMaterial": "LiNi0.8Mn0.1Co0.1O2",
        "anodeMaterial": "Graphite with 5% silicon composite",
        "electrolyte": "LiPF6 in EC/EMC electrolyte",
        "numberOfCells": 52,
        "cellLayout": "13S4P"
      }
    },
    "documentsAndSafety": {
      "usableExtinguishingAgent": {
        "en": "Use large quantities of water for cooling; avoid enclosed-space exposure to vapours.",
        "de": "Mit großen Mengen Wasser kühlen; Dämpfe in geschlossenen Räumen vermeiden."
      },
      "euDeclarationOfConformity": {
        "contentType": "application/pdf",
        "url": "https://documents.example.com/batteries/LMT26B00018427/eu-declaration.pdf",
        "language": "en",
        "resourceTitle": "EU Declaration of Conformity"
      },
      "dismantlingInformation": {
        "contentType": "application/pdf",
        "url": "https://documents.example.com/batteries/LMT26B00018427/dismantling.pdf",
        "language": "en",
        "resourceTitle": "Battery pack dismantling instructions and exploded diagram"
      },
      "safetyMeasures": {
        "en": "Disconnect the battery before servicing. Do not crush, puncture, short-circuit or expose above 60
°C.",
        "de": "Vor Wartungsarbeiten Batterie trennen. Nicht zerdrücken, durchstechen, kurzschließen oder über 60 °C
erhitzen."
      },
      "complianceTestReport": {
        "contentType": "application/pdf",
        "url": "https://documents.example.com/batteries/LMT26B00018427/test-report.pdf",
        "language": "en",
        "resourceTitle": "Performance, durability and safety test report"
```

```
      }
    },
    "currentPerformanceAndStateOfHealth": {
      "ratedCapacity": 18.6,
      "capacityFade": 7.0,
      "power": 925.0,
      "internalResistance": 0.118,
      "remainingCapacity": 18.6,
      "remainingPowerCapability": 925.0,
      "remainingRoundTripEfficiency": 91.3,
      "batteryStatus": "original",
      "chargingDischargingCycleCount": 327,
      "stateOfCharge": 76.4,
      "lastMeasurementTime": "2026-08-27T08:40:00Z"
    }
  }
}
```


9. Full 与 Compressed 的技术差异

对比项	Full	Compressed
结构名称	Fully expanded DataElement representation	Compressed JSON representation
业务字段表示	DataElement 对象数组	普通 JSON Key/Value
是否携带 `objectType`	是	通常不携带
是否携带 `dictionaryReference`	是	通常从外部数据字典解析
是否携带 `valueDataType`	是	由 Schema 或数据字典解析
数据量	较大	较小
可读性	适合通用语义交换、审计、跨系统解释	更符合传统 REST 接口和业务开发习惯
推荐用途	通用 DPP 服务、语义交换、数据治理	App、前端、企业内部服务及高频读取

建议服务端同时支持两种读取格式，共享同一个领域模型和访问控制逻辑，只在序列化层切换表示。默认返回 `compressed`，调用方明确指定 `representation=full` 时返回 full。

10. 字段命名和数据字典实施建议

10.1 分离法规数据点与机器字段

内部数据字典至少保存：

```
{
  "regulatoryDataPointNo": 62,
  "legalLabel": "Remaining capacity",
  "elementId": "remainingCapacity",
  "dictionaryReference": "https://vocabulary.example.com/battery/dp62-remaining-capacity-ah",
  "valueDataType": "xsd:decimal",
  "javaType": "java.math.BigDecimal",
  "unit": "Ah",
  "applicableBatteryCategories": ["LMT", "INDUSTRIAL_IF_APPLICABLE"],
  "legalSource": "Regulation (EU) 2023/1542, Annex XIII 4(b)",
  "mappingStatus": "PROVISIONAL"
}
```

10.2 不要把临时 URI 伪装成欧盟 URI

- 开发阶段使用企业控制的域名或 `urn:company:` 命名空间。
- `.example.com` 仅用于文档示例，生产环境不得使用。
- 欧盟正式语义目录发布后，通过配置或版本化迁移表替换，不改变业务主键和历史数据。

10.3 数值类型

- `xsd:decimal` → Java `BigDecimal`
- `xsd:integer` → Java `BigInteger` 或按范围选择 `Integer/Long`
- `xsd:boolean` → Java `Boolean`
- `xsd:date` → Java `LocalDate`
- `xsd:dateTime` → Java `OffsetDateTime` 或 `Instant`
- `xsd:anyURI` → Java `URI`
- `xsd:base64Binary` → Java `byte[]`

10.4 访问控制

`ReadDPPById` 在序列化 full 或 compressed 之前都必须执行访问控制和字段级过滤。消费者、维修商、再制造商、回收商、市场监管机构可能看到不同字段；不能先返回完整 JSON 再依赖前端隐藏。

11. 当前成熟度说明

欧委会已给出 71 个法规业务数据点，但其 2026-08 指南明确说明，未来更新仍可能补充单个数据点的定义、计量单位和报告格式。因此目前可以完成：

- API 路径、方法、参数、认证和错误响应；
- EN 18223 full/compressed 通用结构；
- 71 个法规数据点的内部字段映射；
- Java DTO、数据库模型和访问控制框架。

仍需保持可配置、等待官方目录进一步确定的内容包括：

- 欧盟官方 `elementId`；
- 正式 `dictionaryReference` URI；
- 部分字段的标准单位、精度和格式；
- 部分枚举及代码表；
- 产品组官方机器可读 Schema 及版本；
- 电池角色访问权限实施法案中的最终字段可见性。